Junru (Andrea) Ai

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RESEARCH INTERESTS

Human-AI interaction; adolescents and generative AI (chatbots, AI companions); emotional reliance and wellbeing in technology use; trust and safety and safety-by-design; qualitative and mixed-methods user research.

EDUCATION

Northwestern University, School of Education and Social Policy — Evanston, IL | Sep 2025 to Sep 2026
M.S. in Technology, People, and Policy (STEM-designated) | GPA: 3.93/4.0

- Relevant coursework: Technology in Context; Technology, Emotion, and Social Relationships; Technology, Psychology, and Education Across Cultures; AI and Data Governance; Data Ethics and Policy; Foundations of Science and Technology Policy; Introduction to Artificial Intelligence; Negotiation and Conflict Management

Wuhan University of Technology — Wuhan, China | Sep 2021 to Jun 2025
B.Econ. in Finance

- Relevant coursework: Statistics, Python Programming, Game Theory, Principles of Management, Corporate Finance

RESEARCH EXPERIENCE

Research Assistant, Culture, Brain, and Human Development Lab, Northwestern University — Mar 2026 to Aug 2026
PI: Prof. Yang Qu

- Completed fMRI training and supported participant recruitment for an adolescent neuroimaging study.
- Pilot-tested in-scanner attention tasks, evaluating their usability and procedural flow from the participant's perspective before data collection.
- Assisted in developing parallel parent and child surveys for a study of family relationships and adolescent development, focusing on how parents and adolescents perceive each other.
- Served as lead research assistant on a three-person search team for a systematic review of AI and digital technologies in K-12 education; identified, retrieved, and archived 265 articles.

Capstone Research Fellow, The Brookings Institution — Dec 2025 to Jun 2026
Supervisor: Dr. Nicol Turner Lee

- Led the talent policy pillar of a comparative study of AI industrial policy in the United States, European Union, and China, analyzing how immigration rules, workforce investment, and retention mechanisms shape national AI competitiveness.
- Co-drafted two policy briefs on AI talent mobility and AI model export controls, later presented as an academic poster.
- Authored a comparative policy blog post, retained by Brookings as internal research material.

SELECTED RESEARCH PAPERS AND PROJECTS

Ai, J. (2026). *Erotic conversations with AI as emotional coping: An interview-based case study.* Unpublished manuscript, Technology, Emotion, and Social Relationships (Prof. Claudia Haase), Northwestern University.

- Designed a semi-structured interview protocol and conducted an in-depth interview with an adult user of AI companion chat on a stigmatized topic.
- Analyzed the case through evolutionary functionalist, social constructionist, and psychoanalytic lenses, identifying a cycle of loneliness, short-term relief, and rebound shame.
- Derived six safety-by-design principles for companion AI, including optional post-interaction aftercare and non-coercive attachment cues.

Ai, J. (2026). *The psychology of influencers and fans: A scoping review of parasocial bonds, celebrity worship, and fan identity in adolescent development.* Unpublished manuscript, Technology, Psychology, and Education Across Cultures (Prof. Yang Qu), Northwestern University.

- Mapped the literature into five conceptual clusters following the Arksey and O'Malley framework and PRISMA-ScR reporting logic.
- Identified three gaps: developmental trajectories, culture as mechanism, and the transition from belonging to risk.

Ai, J., Han, J., Li, L., Zhang, Y., & Yang, Y. (2026). *Delivery riders and algorithmic management: A China-U.S. comparative qualitative study design.* Research proposal, Technology in Context (Prof. Reed Stevens and Vien Nguyen), Northwestern University.

- Co-designed a comparative study of 20 to 25 food-delivery riders using semi-structured interviews, fieldwork, ride-alongs, and app walkthroughs, framed through distributed cognition.
- Developed a coding plan distinguishing how workers interpret algorithmic instructions from how they respond to them.

Ai, J. (2026). *WeChat across two interviews: A comparative interview analysis.* Course project, Technology in Context, Northwestern University.

- Conducted, transcribed, and translated two semi-structured Mandarin interviews; analyzed platform dependence through infrastructural invisibility and context collapse.

Ai, J. (2026). *How a self-checkout system processes a grocery purchase: A distributed cognition field study.* Course project, Technology in Context, Northwestern University.

- Conducted field observation with photo documentation at a supermarket self-checkout; mapped representational transformations and breakdowns, and proposed interface design improvements.

INDUSTRY EXPERIENCE IN AI SAFETY AND TRUST & SAFETY

Content Product Intern, Genius Technology (imoo, children's smart devices) Jun 2024 to Sep 2024

- Analyzed 422 user complaint cases to identify high-frequency content safety risks for child users; authored a remediation report that strengthened risk-handling workflows.
- Collected 1,040 user comments from 230 social media accounts with a Python scraper to assess how families evaluate children's learning devices.

AI Product Intern, StepFun (AI unicorn) Jan 2024 to Mar 2024

- Evaluated AI content moderation services from four cloud vendors (including Volcano Engine and Tencent Cloud) on 1M+ samples across political, sexual, and illegal content; measured false-positive rates and led vendor selection.
- Diagnosed causes of false-positive blocking of user-generated avatars and compared content safety practices across major Chinese LLM products (Kimi, Zhipu, ERNIE Bot, Doubao), recommending refinements to safety labels.

ADDITIONAL PROFESSIONAL EXPERIENCE

Product Operations Intern, Tencent May 2025 to Aug 2025

- Cleaned and analyzed three years of million-record sales data; produced five channel insight reports presented directly to the department head.
- Built a daily-updated sales dashboard connected to the backend system, automatically distributed to the full department.

Project Management Intern, MytePro Dec 2024 to Apr 2025

- Integrated 30,000+ real estate project records across contract, payment, and engineering systems into a unified data ledger.
- Co-authored a master data management manual covering 508 projects and resolved 80+ client technical issues.

PRESENTATIONS

Ai, J., et al. (2026). *Poster on AI talent mobility and AI model export controls.* Poster presented at the M.S. in Technology, People, and Policy Graduation Showcase, Northwestern University.

CONFERENCES ATTENDED

International Association for Safe and Ethical AI (IASEAI '26) | Paris, France | Feb 2026

SKILLS

- **Research methods:** semi-structured interviewing, qualitative coding and thematic analysis, field observation, survey design, usability pilot testing, scoping and systematic literature review, fMRI training
- **Technical:** Python (data analysis, web scraping), SQL, Tableau, Excel
- **Languages:** Mandarin (native), English (fluent; IELTS 7.0)